ShopSmart Retail: Exploratory Data Analysis

Mission DS01 · AnalystLab Africa Experience Lab, Week 1 (Data Science Track)
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1. Executive Summary

ShopSmart earned **₦2.25B** from **16,529 completed orders** in 2025 (average order value ₦135.93K). Customers placed 50,000 orders worth ₦6.82B, but **only one in three was completed**. About ₦4.57B of demand never became revenue. That conversion gap is larger than every other effect in the data put together.

Performance is **remarkably even** across stores, categories, customer segments and months. The best store (Lagos (STR003), ₦102.20M) earns only 26% more than the weakest (Kano (STR002), ₦81.07M), and the gap comes from its higher order completion rate. Profitability is the main concern: the gross margin is 45%, but **27% of completed order lines were sold below cost**, losing ₦225.72M. In addition, **₦3.17B** is tied up in stock, more than a full year of revenue.

The data is complete (no missing values) but **inconsistent**. The data-quality issues below must be fixed at source before ShopSmart builds predictive models.

KPI	Value	KPI	Value
Total revenue (completed)	₦2.25B	Total orders (all statuses)	50,000
Completed orders	16,529	Average order value	₦135.93K
Customers (registered / active)	5,000 / 4,795	Average selling price per unit	₦45.45K
Products / Stores	500 / 25	Units sold (completed)	49,432
Return rate	4.88%	Inventory value (at cost)	₦3.17B

2. Dataset Description

Nine related CSV tables and a data dictionary were provided. Orders is the central fact table, and the other tables describe who bought (Customers), what (Products, through Suppliers), where (Stores, Employees), what happened afterwards (Returns) and what is on the shelf (Inventory).

Table	Rows	Primary key	Contents
Customers	5,000	Customer_ID	Demographics, location, membership segment
Products	500	Product_ID	Name, category, brand, cost and selling price
Orders	50,000	Order_ID	One product line per order: date, qty, price, discount, status
Order_Items	50,000	Order_Item_ID	Duplicate of Orders plus line_total (1:1)
Stores	25	Store_ID	Store name, region, state, city, opening date
Returns	2,500	Return_ID	Returned order, date, reason, refund amount
Employees	125	Employee_ID	Role, store, hire date, salary
Suppliers	489	Supplier_ID	Supplier name, contact, country
Inventory	500	Inventory_ID	Stock on hand and reorder level per product/store

Relationships: Orders → Customers, Products, Stores · Returns → Orders · Inventory → Products, Stores · Employees → Stores · Products → Suppliers (by brand name, since there is no key).

3. Data Cleaning Process

Every step is logged in Cleaned_Data/cleaning_log.csv. The raw files were never modified.

Issue found	Evidence	Action
Column names inconsistent	Mixed case (Customer_ID)	Renamed to snake_case
Duplicates	0 duplicate rows or keys; 56 orders returned more than once	Kept the first return per order (2,500 → 2,442)
Missing values	None in any table	Documented a policy for future extracts
Dates stored as text	M/D/YYYY strings	Parsed to datetime, 0 failures
Discount misapplied	Values 0–20 are percentages but line_total subtracted them as ₦	Recomputed gross, discount and net sales
Order_Items duplicates Orders	Row-for-row identical	Used Orders as the single fact table
Store names not unique	25 stores share 10 names	Added a unique store_label
Region contradicts State	Lagos stores tagged North, South, East and West	Added geo_zone (Nigeria's six zones) from the state
Impossible returns	50% dated before their order; 12% refund more than the sale	Flagged, not deleted
Price inconsistencies	Order prices never match the catalogue; 24% below cost	Flagged below_cost (Part 7)
No product → supplier key	Supplier name = Brand + ' Supplies'	Linked by name (100% match)
Constant column	sub_category = 'General' for all	Dropped

4. Exploratory Findings

Seasonality. Monthly revenue is fairly flat (₦171M–₦201M). **Dec** is the peak (₦201.04M), then October, and Q4 is the strongest quarter. February is lowest, but only because it is short.

Categories. Beauty earns the most (23%), then Groceries and Electronics (about 20% each). Revenue per product is almost the same in every category, so Beauty leads on range size, not on stronger sellers. From Q1 to Q4, Fashion grew +10% and Beauty +8%, while Groceries was flat (+1%).

Stores and regions. Lagos (STR003) leads and Kano (STR002) trails. Every store receives about 2,000 orders, so the stores differ in how many orders they complete (31–36%). Per store, the South West zone performs best and the North West worst, a gap of about 8%.

Customers. Gold, Silver and Bronze members spend within 0.4% of each other per customer, and age, gender, channel and payment method make little difference. **Purchase frequency** is what separates high-value customers (r = 0.79 between completed orders and revenue).

Returns. The return rate is 4.6–5.1% in every category. Only 12 of 500 products have a significantly high rate (top: Home Item 263, 11.6%). Late delivery and wrong item cause 51% of returns, and both are operational failures ShopSmart can control.

Profit and inventory. The top 10 products by revenue and by profit have no products in common, and 57 products lose money overall. Only 21 inventory items are at or below their reorder level, while 417 of 500 hold more than three times it.

5. Visualisations

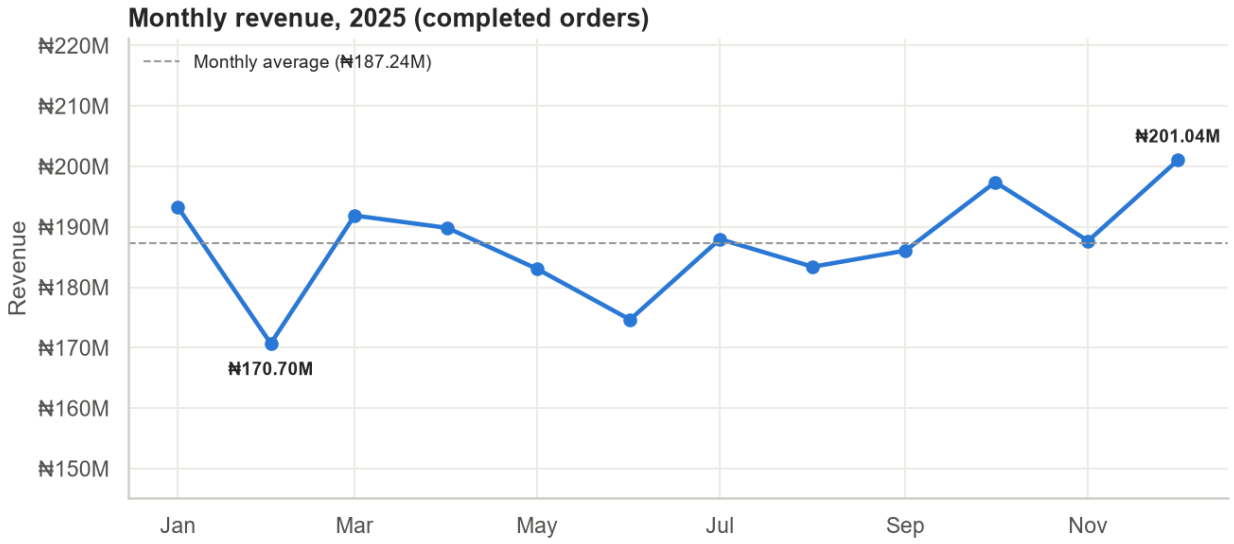


Figure 1. Monthly revenue from completed orders, 2025.

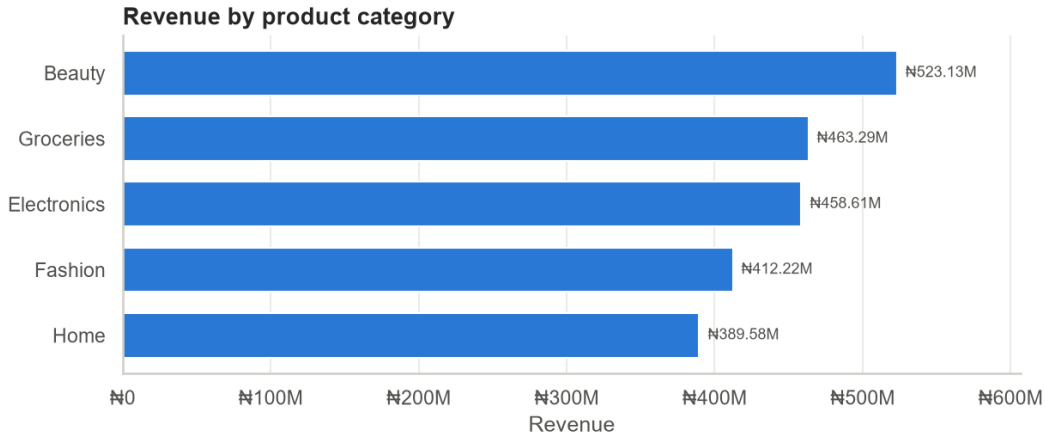


Figure 2. Revenue by category.

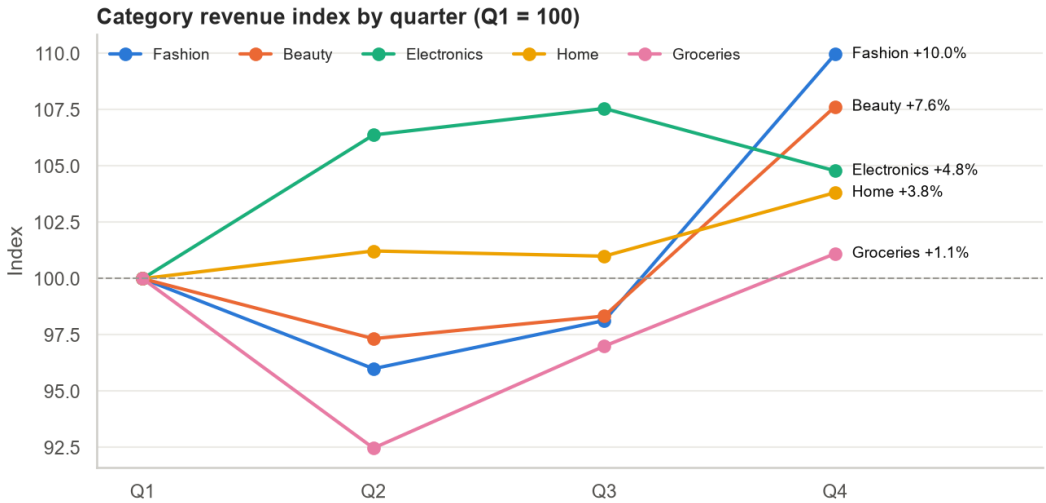


Figure 3. Category revenue indexed to Q1 = 100.

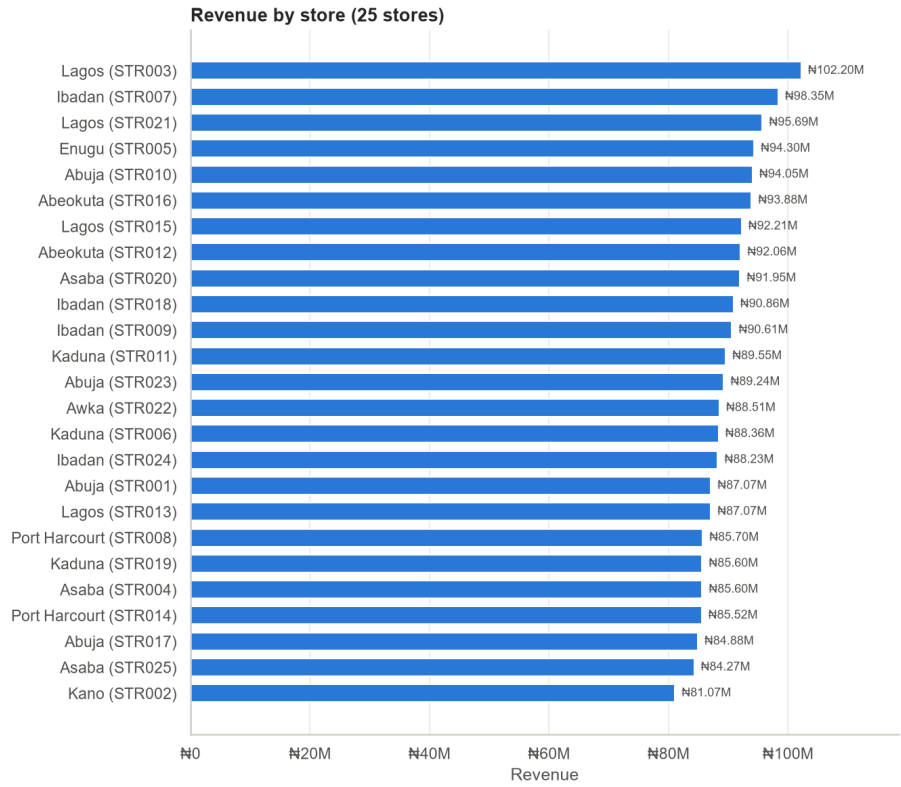


Figure 4. Revenue by store (store labels are unique; names repeat in the source).

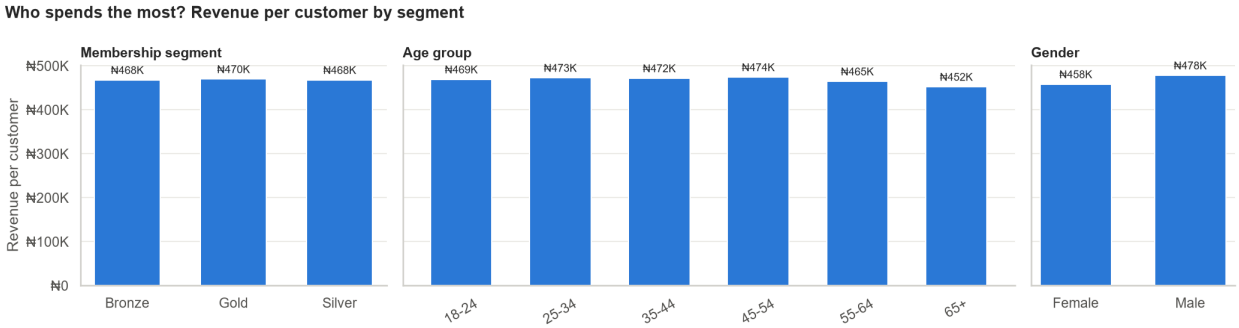


Figure 5. Revenue per customer is flat across membership, age and gender.

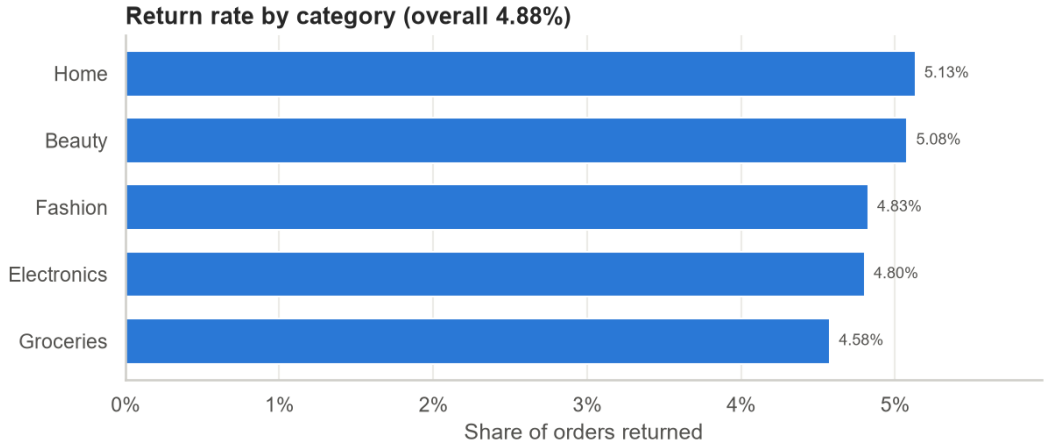


Figure 6. Return rate by category.

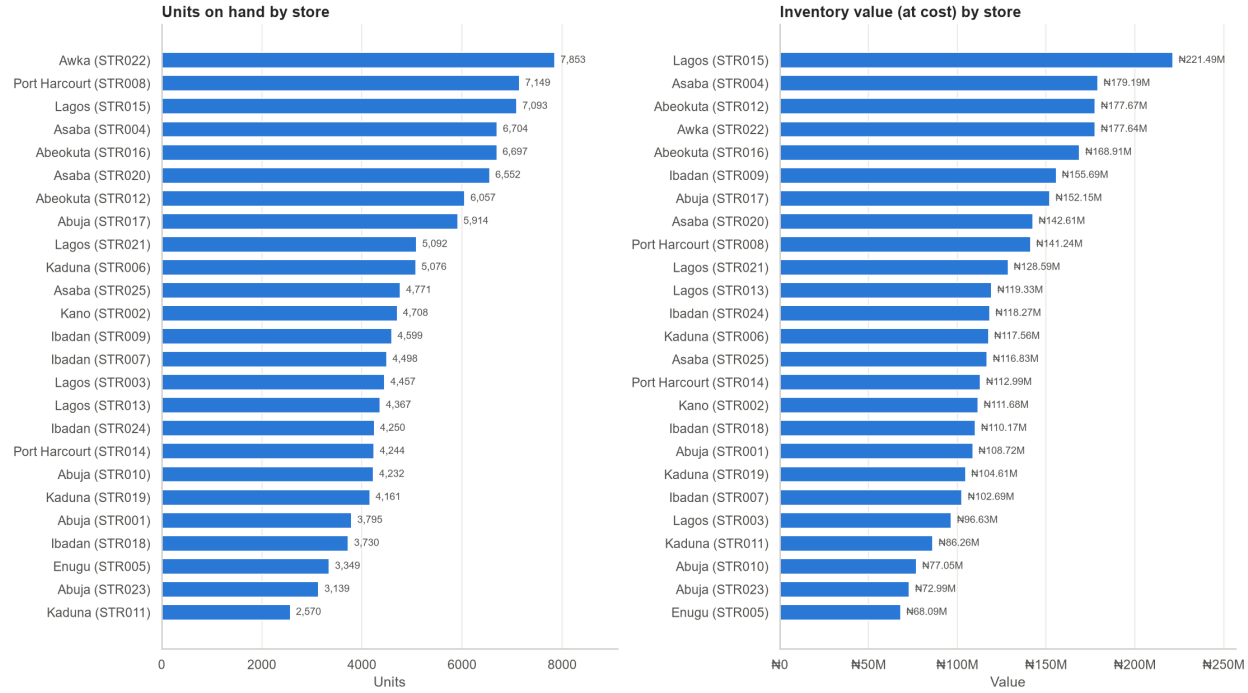


Figure 7. Units and value of stock held per store.

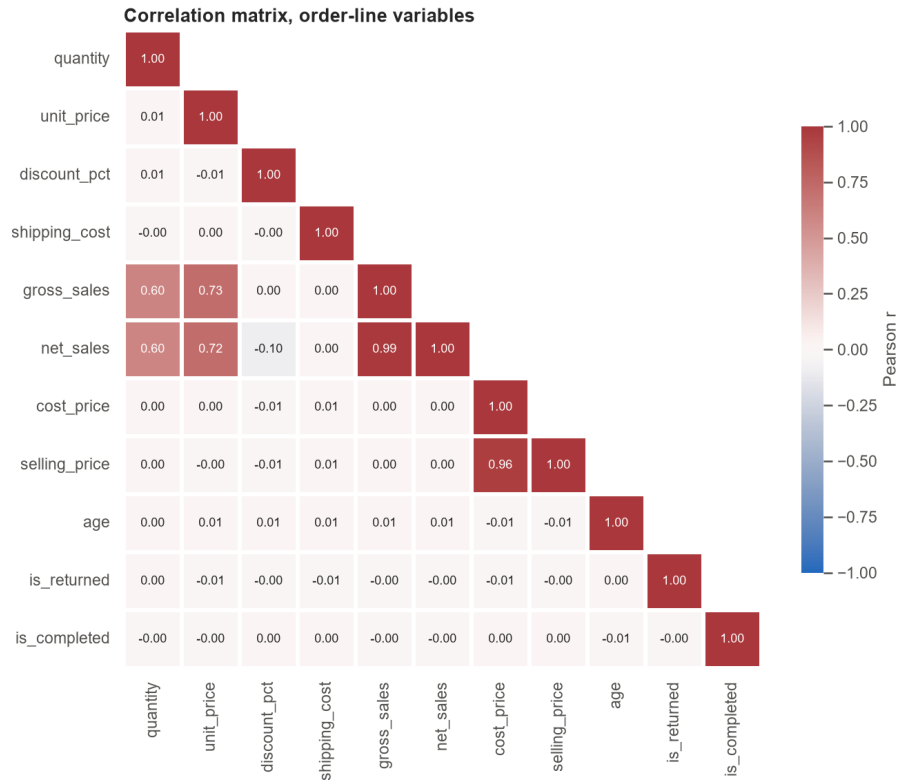


Figure 8. Correlation between order-level variables.

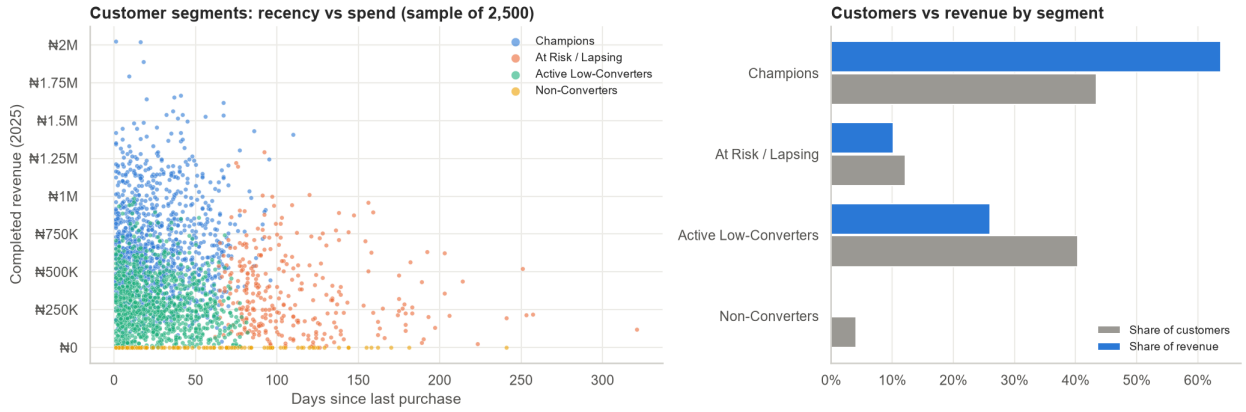


Figure 9. Bonus: K-Means RFM segments.

6. Five Key Insights

1 **Order completion is the biggest lever.** Only 33% of orders complete, and ₦4.57B of demand is lost. The completion rate also explains why some stores beat others.

2 **Revenue does not equal profit.** 27% of completed sales were below cost (₦225.72M lost), and the highest-revenue products are not the most profitable.

3 **Membership tiers do not reflect value.** Gold, Silver and Bronze customers spend the same. Data-driven RFM segments do: 'Champions' are 43% of customers but 64% of revenue.

4 **Inventory is overstocked and badly distributed.** ₦3.17B is held at cost, Lagos (STR015) holds 3× the stock value of Enugu (STR005) despite similar sales, and only 21 items need restocking.

5 **Data quality limits analytics.** Discounts are misapplied, regions contradict states, half the returns are dated before their order, and suppliers cannot be linked by key. These issues must be fixed at source.

7. Five Recommendations

1 **Launch an order-completion programme:** chase pending orders, find out why orders are cancelled (payment, stock, delivery), and copy the practices of the best-converting stores (e.g. Lagos STR003). Raising completion from 33% to 40% would add about ₦470M a year at the current average order value.

2 **Enforce a price floor:** block sales below cost in the POS/e-commerce system and review the 57 loss-making products and 56 loss-making suppliers. Manage the range on profit, not revenue.

3 **Redesign the loyalty programme** around RFM segments: reward Champions, convert Active Low-Converters, win back lapsing customers, and investigate the 205 customers who have never completed an order.

4 **Rebalance inventory:** transfer excess stock to stores with Low items, restock the 21 Low items now, and cut purchase quantities for the overstocked majority to release working capital.

5 **Invest in Beauty and Fashion ahead of Q4,** as the fastest-growing categories with the strongest festive-season uplift. In parallel, fix the source-system data issues and start recording delivery times and supplier lead times.

8. Suggested Machine Learning Problems

Problem	Type	Target / output	Key features & readiness
Order completion prediction	Classification	Completed vs cancelled/pending	Biggest business value. Current features do not separate statuses, so add cart, stock and payment events.
Sales forecasting	Time series	Monthly revenue by store/category	Month, quarter, store, category. Needs 2+ years of history to model seasonality reliably.
Customer churn / CLV	Classification / regression	Lapse in 90 days / 12-month value	Recency, frequency, completion rate, RFM segment. Ready to prototype.
Product recommendation	Collaborative filtering	Next products per customer	10 orders per customer gives purchase histories. Ready to prototype.
Return prediction	Classification	Return_Flag	No current variable correlates with returns. Needs delivery times, reviews and return history.
Inventory forecasting	Regression	Units needed per product/store	Days of supply, stock cover, demand trend. Needs daily stock movements.
Pricing anomaly detection	Anomaly detection	Flag below-cost or off-catalogue prices	Unit price vs cost and catalogue. Ready now, with high immediate value.

Bonus: Customer Segmentation (K-Means)

Customers were clustered on recency, frequency, monetary value and completion rate (log-scaled and standardised, k = 4 chosen by elbow and silhouette = 0.38). The four segments are Champions, Active Low-Converters, At Risk / Lapsing and Non-Converters. Each has the same mix of Gold, Silver and Bronze members, which confirms that the current tiers do not reflect value. Limitations: one year of data, K-Means assumes round clusters, k was partly a business choice, and the data appears synthetic. Re-fit on real data and track segment stability over time.

Limitations of this analysis

- The dataset appears synthetic: distributions are very uniform and names and genders often conflict. Effect sizes are small, and conclusions should be validated on production data.
- Revenue is defined as Completed orders. If Pending orders later complete, revenue will be higher.
- Returns data is unreliable (impossible dates), so return insights are indicative only.
- There is one year of sales, so year-on-year growth and seasonality cannot be confirmed.

Reproducibility: Week1_EDA.ipynb produces every number and chart in this report, and scripts/build_report.py regenerates this PDF from those outputs.